

RESEARCH PAPER

Predictive Forecasting of Care Load and Placement Demand

A Time-Series Forecasting and Capacity-Planning Analysis of the HHS Unaccompanied Alien Children (UAC) Program

Benchmarking Seven Forecasting Models — Statistical and Machine Learning — with Scenario Simulation, Uncertainty Quantification, and Multi-Origin Backtesting

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Abstract

This paper presents an end-to-end forecasting analysis of the U.S. Department of Health & Human Services (HHS) Unaccompanied Alien Children (UAC) Program, using 1,061 days of daily program records (January 26, 2023 – December 21, 2025). The objective is threefold: forecast the number of children in HHS federal care, estimate the future imbalance between intake (transfers) and exits (discharges), and predict short-term discharge demand for placement planning.

Seven forecasting approaches were benchmarked on a strict chronological train/test split — two naive baselines, three statistical models (ARIMA, seasonal ARIMA, and Exponential Smoothing), and two machine-learning ensembles (Random Forest and Gradient Boosting) trained on engineered lag, rolling-window, and calendar features. Random Forest achieved the lowest test-set error (RMSE 6.6, 99.77% accuracy), narrowly ahead of a naive persistence baseline (RMSE 6.9, 99.75%) and clearly ahead of the classical statistical models.

The analysis surfaces a structural finding more consequential than any single model's accuracy score: average daily care load fell from approximately 8,676 children in 2023 to approximately 2,537 in 2025, with daily transfers into care collapsing more than 90% between Q4 2024 and Q2 2025. This regime shift explains why 5-fold rolling-origin cross-validation robustness scores were low (17.2–32.8 out of 100) despite strong single-split accuracy, and why the headline accuracy figures — measured on a 60-day test window with historically unusual stability — should not be read as representative of forecasting difficulty during a volatile or transitional period.

The paper closes with recommendations for shelter capacity planning, discharge-pipeline monitoring, and responsible use of the accompanying forecasting dashboard, along with an explicit discussion of the analysis's limitations.

1. Introduction

1.1 Background

The HHS Office of Refugee Resettlement is responsible for the care and placement of unaccompanied children referred from Customs and Border Protection (CBP) custody. Operational planning for this system depends on anticipating how many children will be in care on a given future date, and whether discharge (placement with a sponsor) capacity will keep pace with intake. Under-forecasting care load risks shelter overcrowding and staffing shortfalls; over-forecasting wastes budget held in reserve capacity that is never used.

1.2 Objectives

- **Primary objective 1:** Forecast the number of children in HHS care for a configurable future horizon (7–90 days).
- **Primary objective 2:** Estimate the future imbalance between intake (Transfers to HHS) and exits (Discharges), i.e., whether the system is gaining or releasing pressure.
- **Primary objective 3:** Predict short-term discharge demand to support placement and staffing decisions.
- **Secondary objective 1:** Provide an early-warning capability for healthcare and shelter planners ahead of capacity breaches.
- **Secondary objective 2:** Quantify forecast uncertainty rather than presenting single-point predictions as certainties.
- **Secondary objective 3:** Compare statistical and machine-learning forecasting approaches on the same data and evaluation protocol.

1.3 Scope and Data Source

This analysis uses a publicly structured daily-record dataset modeled on the HHS UAC Program's reporting format, covering CBP apprehensions and custody, transfers to HHS, HHS care load, and HHS discharges. The work — including the accompanying Streamlit dashboard — is an independent internship case study prepared for portfolio and academic purposes. It is not an official HHS system, is not affiliated with or endorsed by HHS, and should not be used for actual government operational decision-making without independent verification against authoritative HHS data systems.

2. Data and Methodology

2.1 Data Description

The raw dataset contains one row per calendar day across six fields: Date, CBP Apprehended, CBP Custody, Transfers to HHS, HHS Care, and HHS Discharged. After parsing dates, removing thousands-separators from numeric fields, reindexing to a continuous daily frequency, and linearly interpolating the small number of resulting gaps, the working dataset contains 1,061 daily records spanning January 26, 2023 to December 21, 2025.

2.2 Feature Engineering

To give the machine-learning models temporal context that a bare daily figure does not provide, ten engineered features were constructed from the HHS Care series and operational columns:

- Lag features: lag1, lag7, lag14 — the care-load value 1, 7, and 14 days prior
- Rolling statistics: 7-day and 14-day rolling mean, 7-day rolling standard deviation
- Net Pressure: Transfers to HHS minus HHS Discharged (positive = system gaining load)
- Calendar features: day-of-week, month, and a weekend indicator

Four raw operational columns (CBP Apprehended, CBP Custody, Transfers to HHS, HHS Discharged) were included alongside the engineered features, giving 14 total predictors for the machine-learning models.

2.3 Train/Test Split

A chronological (non-random) split was used throughout: the final 60 days (October 23 – December 21, 2025) were held out as a test set, with all preceding days used for training. A chronological split is essential for time-series evaluation — a random split would leak future information into training and produce artificially inflated accuracy.

2.4 Models Benchmarked

- Naive Persistence — tomorrow's value equals today's value; the mandatory sanity-check floor
- Moving Average (7-day) — trailing 7-day mean
- Exponential Smoothing (Holt's linear trend) — classical trend extrapolation
- ARIMA(5,1,2) — classical autoregressive model, no seasonal component
- SARIMA(1,1,1)(1,1,1,7) — seasonal ARIMA with an explicit 7-day seasonal cycle
- Random Forest Regressor (200 trees) — ensemble of decision trees using all 14 features
- Gradient Boosting Regressor (200 trees) — sequential boosted ensemble using all 14 features

2.5 Evaluation Metrics

Each model was scored on four metrics computed against the 60-day test set: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Accuracy (100 – MAPE). RMSE was used as the primary model-selection criterion because it penalizes large individual misses more heavily than MAE — appropriate here, since underestimating a capacity surge is more costly than a small average daily error.

2.6 Additional Validation Methods

- **5-fold rolling-origin cross-validation:** the two ML finalists (Random Forest, Gradient Boosting) were independently re-trained and re-tested across 5 chronological folds (45-day test windows each) to check whether single-split accuracy holds up across different historical periods.
- **Walk-forward multi-origin backtest:** the deployed walk-forward forecasting loop was re-run from approximately 40 independent historical starting points, comparing forecast output against what genuinely happened next, to validate the forecast-uncertainty growth assumption at every horizon day rather than only at Day 1.

3. Exploratory Data Analysis

3.1 Descriptive Statistics

Across the full 1,061-day history, HHS Care load averaged 6,051 children per day (standard deviation 2,843), ranging from a minimum of 1,972 to a maximum of 11,516. This wide spread — a standard deviation equal to

roughly 47% of the mean — is the first indicator that the series is not a single stable regime, a point developed further in Section 3.3.

Variable	Mean	Std. Dev.	Min	Max
CBP Apprehended (daily)	92.8	71.5	0	333
CBP Custody (daily)	170.6	125.4	7	531
Transfers to HHS (daily)	129.1	97.4	0	440
HHS Care (daily)	6,050.8	2,843.5	1,972	11,516
HHS Discharged (daily)	181.2	131.1	0	505
Net Pressure (daily)	-52.0	97.7	-465	206

3.2 Correlation with Care Load

HHS Discharged shows the strongest correlation with HHS Care ($r = 0.933$), followed by Transfers to HHS ($r = 0.719$), CBP Apprehended ($r = 0.703$), and CBP Custody ($r = 0.672$). Net Pressure is negatively correlated ($r = -0.536$), which is mechanically expected: periods of high discharge relative to transfers coincide with a declining or lower care load.

3.3 Structural Decline — The Central EDA Finding

The single most important pattern in this dataset is not a seasonal cycle but a sustained, large-magnitude decline in program volume across the three-year observation window:

Year	Avg. Daily Care Load
2023	8,676
2024	7,020
2025	2,537

Quarterly figures pinpoint the transition precisely. Daily transfers into HHS care averaged 188.1 in Q4 2024, then fell to 36.3 in Q1 2025 and 10.6 in Q2 2025 — a decline of over 90% within two quarters — before stabilizing in the 8–11 range for the rest of 2025. Discharges followed the same pattern (166.0 → 79.6 → 12.1 daily average across the same three quarters). Care load itself fell from an average of 6,133 in Q4 2024 to 2,397 in Q2 2025, subsequently stabilizing near 2,100–2,320.

⚠ Why This Matters More Than Any Model's Accuracy Score

This dataset contains what is effectively two different operational regimes joined end-to-end: a higher-volume period (2023 – Q4 2024, care load roughly 6,000–11,500) and a much smaller, structurally different low-volume period (2025 onward, care load roughly 2,000–2,500).

Any model — however accurate on paper — that is trained across both regimes and evaluated only on the most recent, low-volatility regime risks overstating its real forecasting skill. This is examined directly in Section 5 (Limitations).

3.4 Weekly and Monthly Patterns

Averaged across the full three-year history, HHS Care is mildly higher mid-week (Thursday: 6,109) and lower on Sunday (5,970), a difference of roughly 2.3%. December (7,148) and January (6,951) show the highest monthly averages; July (5,095) the lowest. Because these averages are computed across both operational regimes described in Section 3.3, they should be read as descriptive of the historical average, not necessarily representative of current conditions — a caveat that applies to several full-history statistics in this section.

A time-series decomposition restricted to the most recent 365 days (which falls almost entirely within the 2025 low-volume regime) found a much smaller weekly seasonal amplitude (approximately 14.7 children peak-to-trough) and a residual (unexplained noise) component equal to only 1.7% of total variation — indicating that, within the current regime specifically, trend and weekly seasonality explain the data very well.

3.5 Data Quality Notes

- **Interpolation:** a small number of missing calendar dates were filled via linear interpolation after reindexing to daily frequency; this affects a minority of rows and does not materially change the structural findings above.
- **Outlier detection:** a standard interquartile-range (IQR) check flagged zero outlier days (0.0% of 1,061 records) — not because the data is unusually clean, but because the wide spread introduced by mixing two very different volume regimes inflates the IQR bounds to roughly –5,846 to 16,339, wide enough that no single day falls outside them. This is a useful methodological lesson in its own right: standard outlier-detection thresholds assume a single stable distribution, and can silently fail to flag anything meaningful when the underlying process has shifted regimes rather than simply produced isolated spikes.

4. Model Development and Results

4.1 Full Model Comparison

Model	MAE	RMSE	MAPE (%)	Accuracy (%)
Naive Persistence	6.0	6.9	0.25	99.75
Moving Average (7d)	12.9	14.4	0.55	99.45
SARIMA(1,1,1)(1,1,1,7)	42.3	52.8	1.76	98.24
Gradient Boosting	10.1	13.2	0.43	99.57
ARIMA(5,1,2)	122.4	130.8	5.14	94.86
Exponential Smoothing	191.8	214.4	8.01	91.99
Random Forest (selected)	5.3	6.6	0.23	99.77

Random Forest achieved the lowest RMSE (6.6) and was selected as the production forecasting model. Gradient Boosting was a competitive second among the ML models. The seasonal SARIMA model materially outperformed plain ARIMA (RMSE 52.8 vs. 130.8), confirming that explicitly modeling the 7-day seasonal cycle — rather than leaving it for the autoregressive terms to approximate — adds real value for this series.

4.2 An Important Honest Qualification

Naive Persistence — simply predicting that tomorrow equals today — achieved 99.75% accuracy, within 0.02 percentage points of Random Forest's 99.77%. Random Forest's RMSE improvement over the naive baseline is only 4.3% in relative terms. This is not a failure of the modeling work; it is a direct consequence of the test window's characteristics (Section 4.3) and is reported here because a research paper that only publishes flattering comparisons is not a trustworthy one.

4.3 Why the Test-Set Accuracy Figures Must Be Read in Context

The 60-day test window (October 23 – December 21, 2025) falls entirely within the low-volume, low-volatility regime identified in Section 3.3. Care load during this window ranged narrowly between 2,264 and 2,484 (standard deviation 73.8) — compared to a full-history standard deviation of 2,843.5. In other words, the test window's volatility is roughly 2.6% of the series' historical volatility.

⚠ Interpretation Caution

A model that is simply "close to yesterday" will always score well on a low-volatility window almost by construction. The 99%+ accuracy figures in this section should be understood as measured under unusually favorable, low-volatility conditions — not as a general guarantee of performance during a volatile period or a future regime transition similar to the one identified in Section 3.3.

4.4 Robustness — 5-Fold Rolling-Origin Cross-Validation

To test whether the single-split result generalizes, the two ML finalists were re-evaluated with 5-fold rolling-origin cross-validation (45-day test windows spread across the full history rather than one 60-day window at the end).

Model	Robustness Score (0–100)	Mean CV RMSE	Std. Dev. of CV RMSE	Coefficient of Variation
Random Forest	17.2	36.8	30.5	82.8%
Gradient Boosting	32.8	40.5	27.2	67.2%

Both robustness scores are low. This is directly explained by Section 3.3: because the 5 cross-validation folds are spread across the full 2023–2025 history, some folds land in the high-volume regime and others in the low-volume regime, producing RMSE values that swing substantially between folds (a coefficient of variation of 67–83%, versus the low single-digit variation that would indicate a genuinely stable model). This is treated as a central limitation in Section 6, not glossed over.

4.5 Walk-Forward Multi-Origin Backtest

The standard fan-chart uncertainty visualization used elsewhere in the accompanying dashboard assumes forecast error grows in proportion to the square root of the number of days ahead — a common convention, but one that was validated only against Day-1 errors from the single 60-day holdout. A separate walk-forward backtest, re-running the forecasting loop from roughly 40 independent historical starting points, found that real long-horizon error grows faster than the square-root rule predicts. This means the dashboard's default confidence bands likely understate uncertainty at longer forecast horizons (30+ days) and should be interpreted with additional caution at those horizons specifically.

5. Key Insights

Insight 1 — The program has structurally contracted, not merely fluctuated.

Average daily care load fell roughly 71% from 2023 (8,676) to 2025 (2,537), with the transition concentrated in a two-quarter window (Q4 2024 to Q2 2025) during which daily transfers into care fell over 90%. Any planning exercise using this data should explicitly distinguish "historical average" figures from "current regime" figures, as the two differ by several multiples.

Insight 2 — Random Forest is the strongest available model, but its margin over a naive baseline is thin in the current regime.

Random Forest's feature-rich approach (14 engineered and raw predictors) is structurally better suited to capturing multivariate drivers of care load than any of the single-series statistical models, and it is the only model architecture here compatible with the scenario-simulation capability described in Insight 4. Its advantage over naive persistence specifically, however, is modest under current low-volatility conditions and would need to be re-validated if conditions became more volatile.

Insight 3 — Seasonal structure exists but is secondary to the regime shift.

A real, modest weekly cycle (peaking mid-week) and a modest monthly cycle (peaking in December–January) are both present and were successfully captured by the SARIMA model and by weekly-seasonality features in the ML models. Seasonal ARIMA's large accuracy improvement over plain ARIMA (RMSE 130.8 → 52.8) confirms this. However, the magnitude of seasonal variation is small relative to the magnitude of the 2023–2025 structural decline.

Insight 4 — What-if scenario simulation is possible because Random Forest uses interpretable, adjustable inputs.

Because Random Forest's forecast depends on explicit Transfers-to-HHS and Discharge input features (rather than opaque internal state, as in the statistical models), the accompanying dashboard can simulate four intake/discharge scenarios (Normal, High Transfers, High Discharge, Low Transfers) by adjusting those inputs and re-running the forecast — a capability the statistical models do not support.

Insight 5 — Forecast uncertainty is real and grows faster than the simplest assumption predicts.

The walk-forward backtest (Section 4.5) found that long-horizon forecast error grows faster than a square-root-of-time rule predicts, meaning default confidence intervals likely understate risk at 30+ day horizons. Point forecasts at long horizons should be treated as directional, not precise.

6. Recommendations

6.1 Operational and Planning Recommendations

- **Recalibrate capacity thresholds to current-regime norms.** Static capacity thresholds set using 2023–2024 baseline figures are likely miscalibrated for the current ~2,000–2,500 range; thresholds should be reviewed against the most recent 90–180 days of data, not the full historical average.
- **Monitor the Transfers-to-HHS series as a leading indicator.** Because the 2025 decline was led by a collapse in transfers before care load itself fell, transfers-in should be tracked as an early-warning signal for future regime shifts in either direction.

- **Treat forecasts beyond 30 days as directional, not precise.** Given the confirmed faster-than-vh error growth (Section 4.5), operational commitments (staffing, contracts) based on forecasts should build in wider safety margins the further out the horizon extends.

6.2 Data and Monitoring Recommendations

- **Add explicit regime-change detection.** A rolling change-point or structural-break monitor (e.g., on the 30-day rolling mean of Transfers-to-HHS) would flag a shift like the one identified in Section 3.3 automatically rather than requiring manual EDA to notice it after the fact.
- **Re-run the full model comparison periodically.** Given how much the 2025 regime differs from 2023–2024, model rankings and accuracy figures should be re-validated on a recurring basis (e.g., quarterly) rather than assumed to remain valid indefinitely.

6.3 Modeling and Technical Recommendations

- **Consider regime-aware or regime-segmented models.** A model trained only on the current (2025) regime, or one that explicitly conditions on a regime indicator, would likely produce more stable cross-validation results than a single model spanning both regimes.
- **Widen long-horizon confidence intervals.** Given Section 4.5's finding, the dashboard's fan-chart intervals should be adjusted upward at longer horizons rather than relying on the unadjusted square-root-of-time convention.
- **Report robustness scores alongside accuracy, not instead of it.** Section 4.4's low robustness scores are a more honest signal of real-world reliability than the single-split accuracy figures in Section 4.1, and both should be presented together in any operational use of this model.

7. Limitations

- **Regime mixing in cross-validation.** As shown in Section 4.4, 5-fold CV robustness scores (17.2–32.8) are low because folds are drawn from two structurally different volume regimes. This is the single most important limitation of this analysis and should be resolved before any production use.
- **Test-set accuracy reflects an unusually stable window.** The reported 99%+ accuracy figures (Section 4.3) were measured during a 60-day window with volatility roughly 2.6% of the series' historical volatility, and should not be extrapolated to more volatile conditions.
- **Small effective sample size for a 7-model, cross-validated, scenario-simulated pipeline.** 1,061 daily records is a reasonable amount of data for a single well-chosen model, but is a comparatively small base for the full scope of modeling performed here (7 models, 5-fold CV, 40-origin backtesting). The breadth of this analysis was chosen deliberately for demonstration purposes; a production deployment would likely simplify to fewer, more carefully validated models given the available data volume.
- **Outlier detection method is not well-suited to this series.** As noted in Section 3.5, standard IQR-based outlier detection produced a misleading "zero outliers" result because it assumes a single stable distribution; a regime-aware or change-point-based approach would be more appropriate for a series with a confirmed structural break.
- **Scenario simulation is a sensitivity analysis, not a re-estimated model.** The four what-if scenarios described in Insight 4 adjust Random Forest's input features by a fixed assumption and hold that assumption constant across the forecast horizon; they indicate directional sensitivity to intake/discharge changes, not a formally re-derived probability distribution.

- **Data provenance.** This analysis uses a dataset structured to match the HHS UAC Program's public reporting format for portfolio and educational purposes. It has not been independently verified against an authoritative HHS data system and should not be treated as an official government dataset or forecast.

8. Conclusion

This paper set out to forecast HHS UAC care load, estimate future intake/exit imbalance, and predict short-term discharge demand — and, by every primary and secondary objective stated in Section 1.2, succeeded: Random Forest was identified as the strongest available forecasting model (RMSE 6.6, 99.77% accuracy), uncertainty was quantified through both a standard confidence-interval approach and a more rigorous multi-origin walk-forward backtest, and statistical versus machine-learning approaches were directly compared across seven models.

The more consequential result, however, is the structural one: this program's operational scale changed by roughly a factor of three between 2023 and 2025, concentrated in a two-quarter collapse in intake. No forecasting model, however well-benchmarked, is a substitute for recognizing when the underlying process itself has changed. The recommendations in Section 6 — recalibrated thresholds, leading-indicator monitoring, regime-aware modeling, and periodic re-validation — are offered as the practical response to that finding, and are, in the author's assessment, more actionable for real planners than any single accuracy percentage in Section 4.

— End of Research Paper —